

Permanent Proxies for Bimanual Selection and Manipulation in VR

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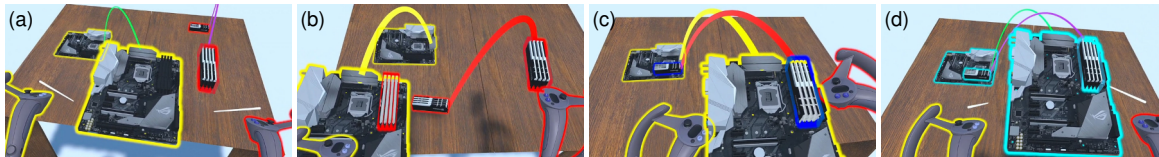


Figure 1: BP technique with permanent proxies (user study condition P). (a) Two previously created proxies for motherboard and RAM. Motherboard proxy and object and left (non-preferred) controller are haloed in yellow, indicating they should be manipulated by non-preferred hand in this user study task. RAM proxy and object and right controller are haloed in red, indicating manipulation by preferred hand. Green and purple arcs connect proxies to objects. (b) Motherboard and RAM proxies are pulled into hands on selection, and their arcs change to yellow and red. Desired 6DoF pose in study task for RAM on motherboard proxy is shown as gray shadow receptacle, haloed in red. (c) RAM proxy and object halos turn blue to indicate RAM proxy has reached desired 6DoF pose. (d) Participant has moved proxies and released them at new 6DoF poses for future use, turning halos cyan.

ABSTRACT

Copies (proxies) of objects are useful for selecting and manipulating objects in virtual reality (VR). Temporary proxies are destroyed after use and must be recreated for reuse. Permanent proxies persist after use for easy reselection, but can cause clutter. To investigate the benefits and drawbacks of permanent and temporary proxies, we conducted a user study in which participants performed 6DoF tasks with proxies in the Voodoo Dolls technique, revealing that permanent proxies were more efficient for hard reselection and preferred.

Index Terms: Virtual reality, interaction techniques.

1 INTRODUCTION

The use of copies (which we refer to as *proxies*) of objects for selection and manipulation in virtual reality (VR) has been widely studied [8, 3, 7, 1]. By manipulating proxies, a user can manipulate the corresponding objects in the virtual environment. Proxies can be either *temporary* (destroyed after use) or *permanent* (persist after use).

One of the best-known methods for manipulating objects through proxies in VR is the Voodoo Dolls technique of Pierce et al. [6], which we refer to as the Bimanual Proxies (BP) technique. With this technique, the user selects a virtual object with their non-preferred hand (object A) and a virtual object with their preferred hand (object B) using image plane techniques [5], creating proxy A' in their nonpreferred hand and proxy B' in their preferred hand. Proxy A' provides an anchor for proxy B': When the user manipulates proxy B' relative to proxy A', object B translates and rotates to follow the manipulation relative to object A. Creating and manipulating proxies makes the BP technique especially powerful for manipulating objects that are inconveniently located.

As implemented by Pierce et al. [6], each proxy is temporary. Although this avoids clutter, if there are objects that are hard to select and need to be manipulated later, it could be advantageous to

keep their proxies from being destroyed when released by making them permanent, as shown in Figure 1. For example, a user might want to alternate between releasing and grabbing a proxy when rotating it to a desired orientation, much like they might release and grab the actual object if it were convenient to manipulate directly. Other examples of objects that are hard to select are objects that are occluded, at a far distance, or appear small.

Pierce briefly suggests [4] the idea of making a proxy permanent by having the user release it by orienting their hand palm up rather than down. Since we are not aware of any evaluation of permanent versus temporary proxies in the BP technique, we present a user study addressing this. We support our results for efficiency using task completion time and preference using subjective responses.

2 RELATED WORK

Virtual proxies were studied in the Worlds in Miniature (WIM) technique [8], in which the user manipulates a small-scale world of proxies representing the full VR world. The proxies corresponding to individual objects in the WIM are permanent. To combat clutter and address scenarios in which objects are vastly different in size, clipping has been implemented for WIMs [9]. Permanent proxies have been used by Oda et al. [3] to allow a remote expert to show a novice user how to complete a physical 6DoF manipulation task with the proxies overlaid in augmented reality. Pohl et al. [7] developed Poros, a system in which users can define a proxy corresponding to a region of space instead of an individual object. All these “proxy spaces” in Poros are permanent. However, we are not aware of any evaluation of permanent versus temporary proxies in techniques involving proxies.

3 SYSTEM DESCRIPTION AND USER STUDY

To analyze the potential benefits and drawbacks of temporary versus permanent proxies, we performed a user study, approved by the Columbia University IRB, to compare two variants of the BP technique. In one variant, all proxies are temporary (Condition T), similar to the original technique of Pierce et al. [6]. In the other variant, all proxies are permanent (Condition P).

We developed a task with a series of steps requiring the participant to perform specific actions (Figure 1). At the start of each step, there is an anchor object with a yellow halo and an object to be moved relative to it with a red halo. The participant must create new proxies or select previously created proxies of these objects, ensuring the proxies of the yellow-haloed and red-haloed objects

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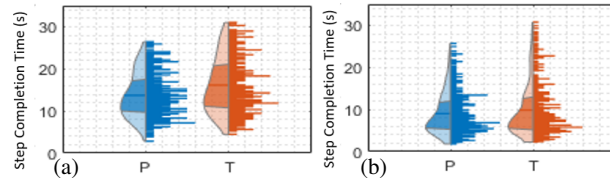


Figure 2: Violin plot of the distribution of step-completion time. (a) Results for hard tasks. (b) Results for easy tasks.

are in their non-preferred and preferred hands, respectively. Then, with the BP technique, the participant can rotate and translate the red-haloed object in 3D relative to the yellow-haloed object, and move it to a specified 6DoF pose to complete the step.

In our study, we distinguish between *hard* and *easy* tasks. Hard selection tasks involve selecting objects that appear small in the participant's FoV ($< 8^\circ$), are occluded by other objects, or are farther ($\geq 10\text{m}$). Easy tasks involve selecting objects that appear large in the participant's field of view (FoV) ($\geq 8^\circ$) and are close ($< 10\text{m}$).

We recruited 15 participants (four female, ages 18–37, $\bar{x} = 24.2$). Five had no prior experience with AR/VR, three had used AR/VR several times, one used VR for short-term projects, two owned a VR headset for gaming, and four used VR for long-term jobs or research. Each received a 15 USD gift card. Participants used a Varjo XR-3 video-see-through AR headset with a 115° horizontal FoV and 90Hz refresh rate. We developed our study in Unity 2022.3.18.

Each participant experienced one block of condition P and one block of condition T, whose order was counterbalanced. Each P and T block consisted of four practice trials, each with four steps, and twelve timed trials, each with six steps. We formulated the following hypotheses: **H1**. *Participants will perform faster with P than with T in hard tasks*; **H2**. *Participants will prefer P over T*. We split the twelve timed trials in each block into six trials with easy tasks and six trials with hard tasks, with the easy trials always given first. Within subject and within condition, we counterbalanced the order of the six easy trials. Similarly, we counterbalanced the order of the six hard trials. After finishing all trials, the participant was asked to fill out a questionnaire that included questions on their demographics, a request to rank the techniques based on their effectiveness, and an unweighted NASA Task Load Index (TLX) survey (as commonly modified to use a 1–7 scale [2]).

3.1 Results

All hypotheses were evaluated with a significance level of $\alpha = .05$. The average step completion time for hard trials was 13.71s under P and 16.14s under T. The distribution can be found in the violin plot of Figure 2(a). Note that the step completion time does not follow a normal distribution, as it was affected by the step difficulty and individual participant performance. We fit a linear mixed-effects model to our dataset. Step completion time served as the observation, while the fixed-effect variable was the condition. (See the model results in the supplementary material for the standard error.) Additionally, the random-effect variables consisted of the step content and the participant. There was no interaction term between random-effect variables. We used P as the baseline, and saw that for T, $p < .0001$, showing participants performed faster in P than T, supporting **H1**. The model has an effect size of $\eta^2 = 0.6618$.

We observed that participants spent more time recreating proxies for hard-to-select objects in T compared to simply reselecting the previously created proxies at earlier steps in P. In addition, some participants told us they were annoyed when they had to release (temporary) proxies in T that were in an awkward pose and recreate them. This is because releasing a temporary proxy that is being manipulated in T requires that a new temporary proxy be created, which may be difficult if the object is hard to select.

We also analyzed the average step completion time for easy trials under P (9.03s) versus T (9.87s) and plotted the results in Fig-

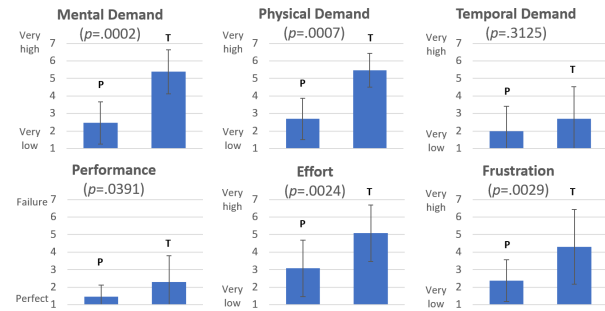


Figure 3: Unweighted NASA TLX results.

ure 2(b). We fit the same linear mixed-effects model on the data from easy trials and found the difference between the completion time for the two conditions is significant ($p = .0003$). The model has an effect size of $\eta^2 = 0.4973$. Note that our easy trials did not involve scenarios with many previously created proxies that would cause clutter. If we tested trials that resulted in cluttered environments with P, we might have found that T performed better than P in these cases. For example, P5 said that T “could be more intuitive if the proxies took up too much space.”

Participants were asked to rank the conditions based on their preferences. All participants preferred P over T, which was significant by a Wilcoxon signed-rank test, with $p = .0002$. Figure 3 shows the raw NASA TLX results for our study (for easy and hard trials combined), with p -values for each metric calculated using Wilcoxon signed-rank tests. The results indicate that the participants found P less mentally demanding, less physically demanding, performing better, requiring less effort, and less frustrating than T, with statistical significance. These findings support **H2**.

4 CONCLUSIONS AND FUTURE WORK

We evaluated the benefits of permanent proxies over temporary proxies when objects are hard to reselect. We found that our users prefer permanent proxies in these cases and perform faster with them. We plan on exploring proxies in other manipulation techniques in a future user study.

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